AUSTRALIAN PROVENANCE PROJECT

ProvenanceOS™

APP | pXRF Field Manual

A simple phone-first workflow for recording Andamooka specimens

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xrf.aosa.au

1. What ProvenanceOS™ is doing

ProvenanceOS™ gives every physical specimen a permanent identity. Everything collected about that specimen is attached to that identity: photographs, fluorescence/afterglow evidence, measurements, XRF results and later observations.

- The database creates the permanent specimen number, you'll never invent a number yourself.
- Original photographs and original Vanta exports are preserved as evidence.
- Derived images, masks and estimates are labelled as derived rather than replacing the original evidence.
- Historical soil/reference data is kept separate from physical specimen records.

2. Before you start

- Use the phone browser to open the ProvenanceOS™ field app.
- Make sure you are signed in and have internet access before starting a session.
- Have specimen bags/labels or another physical labelling method ready.
- Keep the Vanta export file intact. Do not rename, edit or resave the original export before it is imported.

3. Specimen classes

Code	Use when	Meaning
AHM	Andamooka Hard Matrix	Hard matrix material
ACC	Andamooka Coloured Concrete	Coloured concrete / matrix material
ACX	Andamooka Crystal Opal	Crystal opal

OTH	Unknown / Other	Use when classification is uncertain

Important: when you are unsure, use OTH. The permanent specimen ID should not be changed just because the material is later reinterpreted.

4. The field workflow

1. Find a specimen Take the physical stone/bag into the workflow before moving on to the next specimen.

2. Register it Tap + Register specimen. Choose AHM, ACC, ACX or OTH. Add the location and optional notes/weight.

3. Wait for the ID ProvenanceOS™ assigns the permanent number, for example AHM-000027. Do not type or guess the number.

4. Label the physical specimen Write or attach that exact ID to the stone, bag or tray so the physical object and digital record stay linked.

5. Capture visible evidence Take the normal visible-light specimen photograph. Keep the original capture unchanged.

6. Capture special evidence When useful, add fluorescence and/or phosphorescence photos or video. These are separate evidence types.

7. Measurement/dimensions Use the camera measurement tool when a known reference is in the same image plane. Record this as a camera-based estimate, not a laboratory measurement.

8. Attach XRF When a Vanta result is available, attach/import the original export and match its reading to the specimen ID. One export can contain many readings.

9. Finish Check the specimen record. The specimen should show its identity, evidence and any available XRF/measurement data.

> **WHY THE ORDER MATTERS**
>
> The specimen is identified before interpretation. If the chemistry changes our understanding later, the original identity and evidence remain intact.

5. Photographing the specimen

Visible light

- Use a consistent angle where practical.
- Keep enough background/context to understand the capture, but avoid covering the specimen with labels or hands.
- The original phone capture is the evidence. Do not rely only on a cropped or masked version.

Fluorescence / phosphorescence

- Use the correct capture mode in ProvenanceOS™.
- A photo or video can be used.

- Record useful context in notes when known, such as the UV source or capture conditions or the field/area.

6. Measuring with the camera

The camera measurement is a 2D estimate. It needs a known-size reference in the same image plane as the specimen. ProvenanceOS records the reference and calculation details so the estimate can be traced later.

- Do not use a single phone image to claim an accurate thickness measurement.
- Use the measurement as an estimate unless a proper physical measurement has been made.
- If a reference is not available, skip the camera measurement rather than guessing.

7. Vanta / pXRF workflow

> **KEEP THE ORIGINAL EXPORT**
>
> The raw Vanta export is evidence. Do not edit the file before import.

1. Export Move the original Vanta CSV from the instrument to the device/computer used for import.

2. Preserve Keep the file exactly as exported. ProvenanceOS™ stores the original file and a hash for traceability.

3. Import Use XRF Inbox / Import Vanta CSV.

4. Match A reading with an exact ProvenanceOS™ specimen ID is attached to that specimen. Unmatched or conflicting rows are held for review rather than guessed.

5. Review Check the import result before assuming all readings have attached correctly.

6. Interpret Values such as <LOD remain below-detection observations; they are not converted into zero.

Important: missing units are not silently assumed when the source does not provide them. Review the import warning if units are uncertain.

8. What to do when something goes wrong

Situation	What to do
You are not sure what class a stone is	Use OTH. Make a note about the uncertainty.
You accidentally start a second record	Stop. Do not invent a number or try to merge records in the field. Keep the physical specimen identifiable and flag the duplicate for review.
A photo fails to upload	Keep the specimen ID written on the physical item and retry when connected. Do not create a new specimen just because a photo failed.
An XRF reading does not match	Do not guess. Leave it unmatched for review.
The Vanta file has <LOD / ND	Leave it as a below-detection result.
You are unsure whether a value is ppm	Do not alter the value. Check the source/export and review the import warning.
Internet connection drops	Do not assume the record has saved. Keep the physical ID/label and retry when connected.

9. What not to do

- Do not open or edit the Master Excel workbook in the field.
- Do not create specimen IDs manually.
- Do not rename or alter original Vanta exports before import.
- Do not overwrite an earlier result because a later result looks better.
- Do not convert <LOD to zero.
- Do not guess a measurement, sample match or chemistry unit when the evidence is unclear.

10. End-of-day check

- Every physical specimen has a permanent ProvenanceOS™ ID.
- Every labelled specimen can be matched to its digital record.
- Important visible colour / UV / afterglow evidence has been captured.
- Any available XRF files are preserved and imported or clearly marked for later review.
- Anything uncertain has a note or review flag rather than a guess.

THE GOLDEN RULE

When in doubt: preserve the original evidence, keep the specimen identifiable, and flag the uncertainty. We will never make the record look cleaner by guessing.